

FINITE-SUBSET CIRCLE EXPANSION: PARITY-SPLIT FIXED COUNTS AND ALTERNATING ZETA FACTORS

ANONYMOUS

ABSTRACT. Let $m_d(x) = dx \pmod{1}$ on the circle, where $d \geq 2$, and let $H_{d,k}$ send a nonempty subset of at most k circle points to its image under m_d . This gives a nonlinear, cardinality-collapsing map on the k -dimensional finite-subset space. We compute its complete unsigned periodic ledger. If $Q = d^n$, then fixed subsets of exact cardinality j number

$$E_j(Q) = \frac{(Q-1)(Q^j - (-1)^j)}{Q+1}.$$

Summing through $j = k$ yields $Q(Q^k - 1)/(Q + 1)$ for even k and $(Q^{k+1} - 1)/(Q + 1)$ for odd k . The resulting Artin–Mazur zeta function is a finite alternating product of the factors $1 - d^r z$. We also obtain an exact Möbius census and a prime-orbit asymptotic, prove $h_{\text{top}}(H_{d,k}) = k \log d$ through a uniformly finite-to-one product cover, and recover (d, k) from the zeta function. The proof rests on a binary Euler transform: a fixed finite subset is a disjoint union of base cycles, so each base cycle is either absent or present once. Independent coefficient and literal rational-circle enumerations verify the registered finite controls.

1. INTRODUCTION

Finite-subset dynamics retains the motion of individual points while adding collisions: distinct points may acquire the same image, and the cardinality of a configuration may drop. Even for the expanding circle map $x \mapsto dx$, this quotient is not a group endomorphism and is not a fixed Cartesian power. The collision strata interact with periodic counting in a particularly rigid way.

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and let X_k be the space of all nonempty subsets of $\mathbb{T}$ of cardinality at most k . For $d \geq 2$, define

$$H_{d,k} : X_k \longrightarrow X_k, \quad H_{d,k}(A) = \{dx \pmod{1} : x \in A\}.$$

We give an exact unsigned periodic classification of this quotient, with a standard finite-to-one entropy control. Three conclusions carry the main content.

- (1) We classify every fixed configuration of every iterate and derive a bivariate Euler product for the exact-cardinality strata. Its rational collapse gives the closed formula in the abstract.
- (2) The total fixed count has an even–odd split, and the complete Artin–Mazur zeta function is an alternating product of k linear factors (of $k + 1$ factors when k is odd and the factor at d^0 is included). Möbius inversion gives the least-period orbit census and the asymptotic $d^{km}/m + O_{d,k}(d^{(k-1)m}/m)$.
- (3) The quotient map $\mathbb{T}^k \rightarrow X_k$ has uniformly finite fibers. It therefore transfers the product entropy exactly, giving $k \log d$. The two outermost zeta factors recover d and k , so the parameter family is rigid under topological conjugacy.

The first signal is visible without asymptotics. At time n , put $Q = d^n$. The counts for subsets of sizes 1, 2, 3, 4 are

$$Q - 1, \quad (Q - 1)^2, \quad (Q - 1)(Q^2 - Q + 1), \quad (Q - 1)^2(Q^2 + 1).$$

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Their partial sums are $Q - 1$, $Q(Q - 1)$, $(Q - 1)(Q^2 + 1)$, and $Q(Q - 1)(Q^2 + 1)$. The alternating boundary term is the early anomaly that leads to the parity theorem.

2. FINITE-SUBSET SPACES AND THE OWNER BOUNDARY

For a compact metric space Y , its k th finite-subset space is

$$\exp_k(Y) = \{A \subseteq Y : 1 \leq |A| \leq k\},$$

with the Hausdorff topology. Equivalently, it has the quotient topology of

$$\pi_k : Y^k \longrightarrow \exp_k(Y), \quad \pi_k(y_1, \dots, y_k) = \{y_1, \dots, y_k\}.$$

We use $X_k = \exp_k(\mathbb{T})$. Tuffley determined the topology of these circle spaces, their homotopy type, and the degree of maps induced from circle maps [12]. In particular, X_k has topological dimension k ; for $k \geq 4$ it is generally a stratified compactum rather than a manifold.

The notation needs a firewall. Some hyperspace papers call $\exp_k(Y)$ an “ n -fold symmetric product.” The topological symmetric power $\mathrm{SP}^k(Y) = Y^k/S_k$ remembers multiplicities, whereas $\exp_k(Y)$ forgets them. The two agree for $k \leq 2$ but differ for $k \geq 3$: for example, (a, a, b) and (a, b, b) define different points in $\mathrm{SP}^3(Y)$ and the same point in $\exp_3(Y)$.

General induced hyperspace dynamics is well developed. Akin, Auslander, and Nagar study the relation between a map and the induced map on compact subsets [1]. Higuera and Illanes record, among other facts, the finite-permutation mechanism behind periodic induced subsets [8]. Gómez-Rueda, Illanes, and Méndez prove qualitative transfer statements for periodicity, recurrence, and related properties on finite-subset spaces [7]. Fernández, Good, and Puljiz analyze admissible period sets in hyperspaces and symmetric products [5]. These results supply the qualitative owner boundary; they do not give the unsigned exact-cardinality ledger computed below.

For multiplication maps of the circle, Tan studies rotational finite subsets and their rotation data [11]. Our count instead includes every union of base cycles that fits below a cardinality cutoff, without imposing one cyclic order type or a rotation number.

Finally, Artin–Mazur counting records cardinalities of fixed sets [2]. Lefschetz zeta functions and symmetric-power identities record fixed point indices; recent functorial treatments include [4, 6]. We compare the two ledgers in section 6, but do not identify them. The residual theorem package here is the circle-multiplication specialization and rational collapse of binary cycle selection, its parity split, the resulting unsigned Artin–Mazur factors, temporal census, and parameter rigidity. The finite-to-one entropy argument is a standard application of Bowen’s factor inequality, not a claimed family-specific mechanism. A bounded search through 28 August 2026 did not locate that combined package. This is not a priority proof, and external novelty language remains on hold.

3. INVARIANT FINITE SETS AS BINARY CYCLE SELECTIONS

Fix $d \geq 2$ and $k \geq 1$. At time $n \geq 1$, write

$$Q = d^n, \quad g_Q = m_d^n = m_Q : \mathbb{T} \longrightarrow \mathbb{T}.$$

For $\ell \geq 1$, let $O_\ell(Q)$ denote the number of g_Q -orbits of least period ℓ .

Lemma 3.1 (base orbit inventory). *For every $\ell \geq 1$,*

$$(1) \quad |\mathrm{Fix}(g_Q^\ell)| = Q^\ell - 1, \quad O_\ell(Q) = \frac{1}{\ell} \sum_{e|\ell} \mu\left(\frac{\ell}{e}\right) (Q^e - 1).$$

Moreover, as a formal power series,

$$(2) \quad \prod_{\ell \geq 1} (1 - u^\ell)^{-O_\ell(Q)} = \exp\left(\sum_{r \geq 1} \frac{Q^r - 1}{r} u^r\right) = \frac{1 - u}{1 - Qu}.$$

Proof. The equation $g_Q^\ell(x) = x$ is $(Q^\ell - 1)x = 0$ in $\mathbb{T}$. Its solutions form the cyclic subgroup of order $Q^\ell - 1$. If $R_\ell(Q)$ is the number of points of least g_Q -period ℓ , then $Q^\ell - 1 = \sum_{e|\ell} R_e(Q)$. Möbius inversion gives $R_\ell(Q) = \ell O_\ell(Q)$ and proves (1).

The infinite product in (2) is well defined formally, because only finitely many factors affect each coefficient. Its logarithm has coefficient

$$[u^r] \sum_{\ell \geq 1} O_\ell(Q) \sum_{a \geq 1} \frac{u^{a\ell}}{a} = \frac{1}{r} \sum_{\ell|r} \ell O_\ell(Q) = \frac{Q^r - 1}{r}.$$

Exponentiating gives the first equality. Summing the two geometric logarithms gives the rational expression. $\square$

We denote the common formal series in (2) by $\zeta_{g_Q}(u)$.

The next lemma is the point where the induced dynamics becomes a binary Euler transform. The finite-permutation observation is general and appears in the induced finite-subset literature; see, for example, [8, Lemma 4.7]. We include the short proof to fix the exact orbit inventory used below.

Lemma 3.2 (orbit-union classification). *Let $A \subset \mathbb{T}$ be nonempty and finite. Then $g_Q(A) = A$ if and only if A is a disjoint union of complete finite g_Q -orbits. This decomposition is unique.*

Proof. If $g_Q(A) = A$, the restriction $g_Q|_A : A \rightarrow A$ is surjective. A surjection of a finite set is bijective, so $g_Q|_A$ is a permutation. Its permutation cycles are complete g_Q -orbits in $\mathbb{T}$, they are pairwise disjoint, and their union is A . The cycle decomposition of a permutation is unique. Conversely, g_Q maps every complete orbit onto itself, so it maps any finite disjoint union of such orbits onto that union. $\square$

For $j \geq 1$, define the exact-cardinality fixed stratum

$$E_j(Q) = |\{A \in X_j : |A| = j, g_Q(A) = A\}|.$$

The number is independent of the ambient cutoff once that cutoff is at least j . These strata are used only to partition fixed points; the exact- j stratum is not generally forward invariant because collisions may reduce cardinality.

Theorem 3.3 (binary Euler quotient). *As a formal power series in u ,*

$$(3) \quad \Phi_Q(u) := 1 + \sum_{j \geq 1} E_j(Q) u^j = \prod_{\ell \geq 1} (1 + u^\ell)^{O_\ell(Q)} = \frac{\zeta_{g_Q}(u)}{\zeta_{g_Q}(u^2)} = \frac{1 - Qu^2}{(1 - Qu)(1 + u)}.$$

Consequently, for every $j \geq 1$,

$$(4) \quad E_j(Q) = \frac{(Q - 1)(Q^j - (-1)^j)}{Q + 1}.$$

Proof. By lemma 3.2, a fixed finite subset is specified by deciding, for each g_Q -orbit, whether to include that orbit. An orbit of length ℓ contributes either 1 or u^ℓ , and there are $O_\ell(Q)$ such orbits. Multiplying these independent binary choices proves the first product.

Since $1 + u^\ell = (1 - u^{2\ell})/(1 - u^\ell)$, lemma 3.1 gives

$$\prod_{\ell \geq 1} (1 + u^\ell)^{O_\ell(Q)} = \frac{\prod_{\ell} (1 - u^{2\ell})^{O_\ell(Q)}}{\prod_{\ell} (1 - u^\ell)^{O_\ell(Q)}} = \frac{\zeta_{g_Q}(u)}{\zeta_{g_Q}(u^2)}.$$

Substitution of (2) cancels $1 - u$ and produces the final rational function in (3).

For coefficient extraction, rewrite it as

$$\Phi_Q(u) = 1 + \frac{(Q - 1)u}{(1 - Qu)(1 + u)}.$$

The coefficient of u^{j-1} in the reciprocal product is

$$\frac{Q}{Q + 1} Q^{j-1} + \frac{1}{Q + 1} (-1)^{j-1} = \frac{Q^j - (-1)^j}{Q + 1}.$$

Multiplication by $Q - 1$ proves (4). The quotient is an integer because $Q^j - (-1)^j$ is divisible by $Q + 1$; integrality also follows from the orbit-selection interpretation. $\square$

4. PARITY-SPLIT FIXED COUNTS AND ZETA FACTORS

Put

$$A_k(Q) = \sum_{j=1}^k E_j(Q).$$

Since $H_{d,k}^n(A) = g_Q(A)$ for $Q = d^n$, this is exactly $|\text{Fix}(H_{d,k}^n)|$.

Theorem 4.1 (parity collapse). *For $Q \geq 2$ and $k \geq 1$,*

$$(5) \quad A_k(Q) = \begin{cases} \frac{Q(Q^k - 1)}{Q + 1}, & k \text{ even}, \\ \frac{Q^{k+1} - 1}{Q + 1}, & k \text{ odd}. \end{cases}$$

Equivalently, with

$$r_0(k) = \begin{cases} 1, & k \text{ even}, \\ 0, & k \text{ odd}, \end{cases}$$

one has the alternating polynomial

$$(6) \quad A_k(Q) = \sum_{r=r_0(k)}^k (-1)^{k-r} Q^r.$$

Proof. Summing (4) gives

$$A_k(Q) = \frac{Q - 1}{Q + 1} \left(\frac{Q(Q^k - 1)}{Q - 1} - \sum_{j=1}^k (-1)^j \right).$$

The alternating sum is zero for even k and -1 for odd k . This proves (5). Polynomial division by $Q + 1$ gives (6); its lower endpoint is Q^1 in the even case and Q^0 in the odd case. $\square$

Define the Artin–Mazur zeta function formally by

$$\zeta_{d,k}^{\text{AM}}(z) = \exp \left(\sum_{n \geq 1} |\text{Fix}(H_{d,k}^n)| \frac{z^n}{n} \right).$$

Theorem 4.2 (alternating Artin–Mazur factors). *For $d \geq 2$ and $k \geq 1$,*

$$(7) \quad \boxed{\zeta_{d,k}^{\text{AM}}(z) = \prod_{r=r_0(k)}^k (1 - d^r z)^{(-1)^{k-r+1}}}.$$

In particular,

k	$\zeta_{d,k}^{\text{AM}}(z)$
1	$\frac{1 - z}{1 - dz}$
2	$\frac{1 - dz}{1 - d^2 z}$
3	$\frac{(1 - z)(1 - d^2 z)}{(1 - dz)(1 - d^3 z)}$
4	$\frac{(1 - dz)(1 - d^3 z)}{(1 - d^2 z)(1 - d^4 z)}$

Proof. Insert $Q = d^n$ into (6). Then

$$\begin{aligned} \log \zeta_{d,k}^{\text{AM}}(z) &= \sum_{r=r_0(k)}^k (-1)^{k-r} \sum_{n \geq 1} \frac{(d^r z)^n}{n} \\ &= - \sum_{r=r_0(k)}^k (-1)^{k-r} \log(1 - d^r z). \end{aligned}$$

Exponentiation gives (7). The displayed examples are direct specializations. $\square$

Corollary 4.3 (parameter rigidity). *Within the family $d \geq 2$, $k \geq 1$, the Artin–Mazur zeta function determines the ordered pair (d, k) . Hence $H_{d,k}$ and $H_{d',k'}$ can be topologically conjugate only if $(d, k) = (d', k')$.*

Proof. The factorization (7) has no internal cancellation, because d^r is strictly increasing in r . Its positive pole nearest the origin is d^{-k} , and its positive zero nearest the origin is $d^{-(k-1)}$. For $k = 1$ this zero is the factor at $d^0 = 1$. The ratio of the zero location to the pole location is d . The pole then recovers k . Topological conjugacy bijects the fixed sets of corresponding iterates and therefore preserves the Artin–Mazur zeta function. $\square$

5. LEAST-PERIOD ORBITS

Let $P_{d,k}(m)$ be the number of $H_{d,k}$ -orbits of least temporal period m .

Theorem 5.1 (exact temporal census). *For every $m \geq 1$,*

$$(8) \quad P_{d,k}(m) = \frac{1}{m} \sum_{e|m} \mu\left(\frac{m}{e}\right) A_k(d^e),$$

where A_k is given by either (5) or (6). This expression is a nonnegative integer. For fixed $d \geq 2$ and $k \geq 2$,

$$(9) \quad P_{d,k}(m) = \frac{d^{km}}{m} + O_{d,k}\left(\frac{d^{(k-1)m}}{m}\right).$$

Proof. Let $R_{d,k}(m)$ count points of least $H_{d,k}$ -period m . Every point fixed by the n th iterate has a unique least period dividing n , so

$$A_k(d^n) = \sum_{m|n} R_{d,k}(m).$$

Möbius inversion gives the numerator of (8). Every least- m orbit contains exactly m points, hence $P_{d,k}(m) = R_{d,k}(m)/m$. This proves both the formula and its nonnegative integrality without an auxiliary congruence argument.

Equation (6) gives

$$A_k(d^m) = d^{km} + O_{d,k}(d^{(k-1)m}).$$

Every proper divisor e of m satisfies $e \leq m/2$. Also $A_k(d^e) \leq C_{d,k} d^{ke}$, and therefore

$$\sum_{\substack{e|m \\ e < m}} A_k(d^e) \leq C_{d,k} \sum_{e=1}^{\lfloor m/2 \rfloor} d^{ke} = O_{d,k}(d^{km/2}) = O_{d,k}(d^{(k-1)m}),$$

where the last inequality uses $k \geq 2$. Substitution in (8) proves (9). $\square$

6. ENTROPY AND TWO OWNER CONTROLS

The finite-subset quotient is not a product, but it has a product cover with uniformly bounded fibers. This section records a general entropy control, not a residual novelty claim for the circle family; related entropy questions for induced hyperspace maps are surveyed and developed in [9].

Theorem 6.1 (finite-to-one entropy equality). *For $d \geq 2$ and $k \geq 1$,*

$$(10) \quad h_{\text{top}}(H_{d,k}) = k \log d.$$

Proof. Let $M_{d,k} = m_d^{\times k}$ on $\mathbb{T}^k$. The quotient map π_k satisfies

$$\pi_k \circ M_{d,k} = H_{d,k} \circ \pi_k.$$

It is onto. If $A \in X_k$ has j elements, then $\pi_k^{-1}(A)$ consists of the k -tuples in which every member of A occurs at least once. In particular,

$$|\pi_k^{-1}(A)| \leq j^k \leq k^k,$$

uniformly in A .

Entropy does not increase under a factor, so $h_{\text{top}}(H_{d,k}) \leq h_{\text{top}}(M_{d,k})$. Bowen's factor inequality bounds the reverse difference by the supremum of the Bowen entropies of the fibers [3]. Every fiber above has at most k^k points, independent of orbit length, and hence has Bowen entropy zero. Thus the two entropies are equal. This is the standard uniformly finite-to-one consequence of Bowen's factor inequality. The product formula and $h_{\text{top}}(m_d) = \log d$ give $h_{\text{top}}(M_{d,k}) = k \log d$. $\square$

Homological control. Tuffley's results [12] give

$$X_k \simeq S^{2\lceil k/2 \rceil - 1}, \quad \deg(H_{d,k}) = d^{\lceil k/2 \rceil} \quad \text{on top nonzero homology.}$$

Since this homology dimension is odd, the Lefschetz number of the n th iterate is

$$L(H_{d,k}^n) = 1 - d^{n\lceil k/2 \rceil},$$

and the corresponding Lefschetz zeta function is

$$\zeta_{d,k}^L(z) = \exp\left(\sum_{n \geq 1} L(H_{d,k}^n) \frac{z^n}{n}\right) = \frac{1 - d^{\lceil k/2 \rceil} z}{1 - z}.$$

For $k \geq 2$, this homological scale differs from the unsigned growth scale d^{kn} in [theorem 4.1](#). Thus the Artin–Mazur factors in (7) are not a reformulation of the Lefschetz or fixed-index formulas owned by the cited literature.

Multiplicity control. Periodic points of symmetric-product mappings have a direct earlier literature, including Rallis [10]; the following calculation is only a comparison control. On the genuine symmetric power $\text{SP}^k(\mathbb{T})$, equality of a multiset with its image first forces its finite support to map onto itself. Thus g_Q permutes the support, and the multiplicity is constant along every support cycle. A fixed multiset may consequently use a complete g_Q -orbit with any nonnegative common multiplicity. Therefore its mass generating function is

$$\prod_{\ell \geq 1} (1 - u^\ell)^{-O_\ell(Q)} = \frac{1 - u}{1 - Qu}.$$

The coefficient at mass $k \geq 1$ is $Q^{k-1}(Q - 1)$. Hence the induced map on $\text{SP}^k(\mathbb{T})$ has

$$\zeta_{\text{SP}^k(m_d)}^{\text{AM}}(z) = \frac{1 - d^{k-1}z}{1 - d^k z}.$$

This agrees with the finite-subset result at $k \leq 2$, when multiplicity and support encode the same quotient, but it lacks the lower alternating factors for $k \geq 3$. The comparison isolates the mechanism: arbitrary orbit multiplicity gives the ordinary Euler transform, whereas forgetting multiplicity leaves a binary transform and creates the parity boundary.

7. EXACT CONTROLS, BOUNDARY CASES, AND SCOPE

The registered script implements two independent finite probes. The first computes the Möbius orbit counts $O_\ell(Q)$ and multiplies the truncated binary Euler factors with exact integer arithmetic. For $2 \leq Q \leq 8$ and $j, k \leq 9$, it compares every coefficient and partial sum with (4) and (5). A separate formal-factor probe checks every logarithmic coefficient and the outer pole/zero recovery for $2 \leq d \leq 7$, $1 \leq k \leq 9$, and iterates through 15. It also checks the temporal Möbius census, including $k = 1$, and its divisibility.

The second probe works with actual rational circle points. For a chosen (Q, k) it takes

$$L = \text{lcm}_{1 \leq \ell \leq k}(Q^\ell - 1),$$

decomposes multiplication by Q on $\mathbb{Z}/L\mathbb{Z}$ into literal cycles, and counts cycle selections by cardinality. Small cases additionally enumerate every subset of $\mathbb{Z}/L\mathbb{Z}$ of size at most k and test $QA = A$ directly. The two routes share neither coefficient extraction nor subset enumeration.

The assumptions are material. We exclude $d = 1$, because the identity has uncountably many fixed configurations and its unsigned Artin–Mazur zeta function is not defined by finite cardinalities. The case $k = 1$ is included as a base control, but $k \geq 2$ is the genuinely higher-dimensional family. We exclude the empty subset; adding it would add one fixed point to every iterate and multiply (7) by $(1 - z)^{-1}$. Exact-cardinality coefficients $E_j(Q)$ are fixed-point strata, not zeta functions of invariant exact- j subsystems.

This manuscript is an internal theorem record. Public posting, submission, and absolute priority language remain on external hold pending a broader specialist and database search.

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